

CYBER PHYSICAL SYSTEMS - CAR RACING

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2022

Gym Car Racing

Driving the car around a street track



Requirements

If we define $x(t)$ = position of car at time t , y = the car is on track,
 z = The grass is in the visible portion of the environment

Hard Requirements:

- ▶ Requirement H1: The car always needs to be moving:
 $G(x(t) \neq x(t + 1))$
- ▶ Requirement H2: We always need to see the track : $G(z)$
- ▶ Requirement H3: Eventually we always need to go back to the track: $G(F(y))$

Soft Requirements:

- ▶ Requirement S: We always have to be on track: $G(y)$

Robustness

- ▶ Distance from grass - distance from street

Model Predictive Control

- ▶ Generate random trajectories
- ▶ Learn to predict the system
- ▶ Based on the system prediction, choose action that maximizes predicted robustness on a time horizon

Model Predictive Control

Algorithm 1 Model Predictive Control using F system predictor

for Each time step t in episode **do**

for a_t in possible actions **do**

for $i \in \{0, \dots, \tau\}$ **do**

$\hat{s}_{t+i+1} = F(\hat{s}_{t+i}, \hat{a}_{t+i})$, choose a_{t+i+1} as the one that
minimizes the cost $\rho(\hat{s}_{t+i+1})$

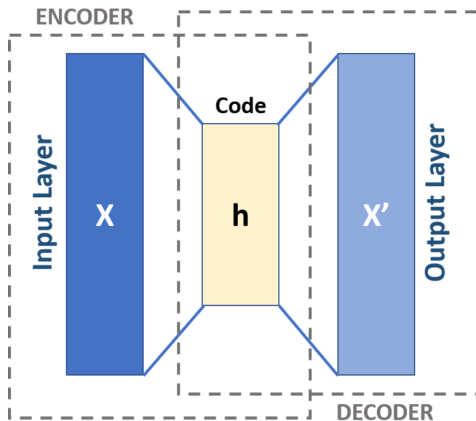
end for

end for

 Execute first action a^* from the sequence $(a_t, \dots, a_{t+\tau})$ that
minimizes the total cost

end for

Auto Encoder Neural Network



Training

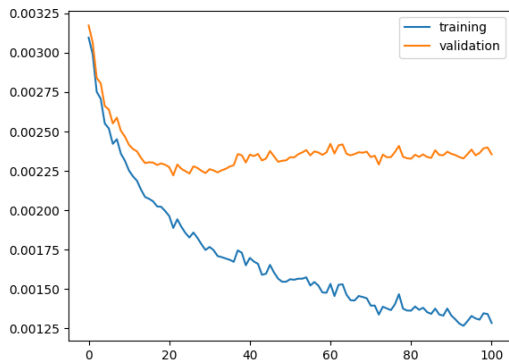


Figure: Loss during training

Control Time Horizon=15

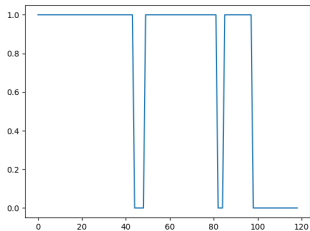


Figure: Car is on street

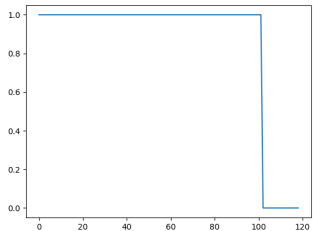
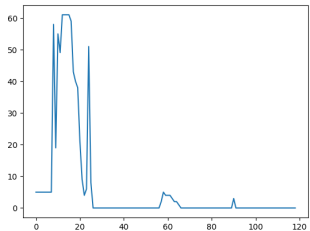
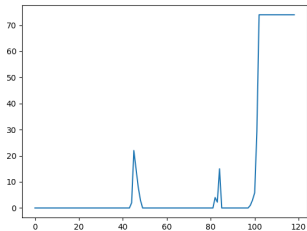


Figure: Street is viewable



Control Time Horizon=20

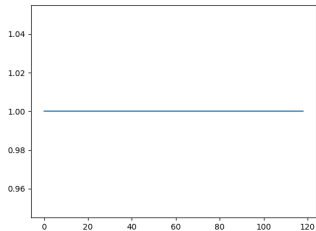


Figure: Car is on street

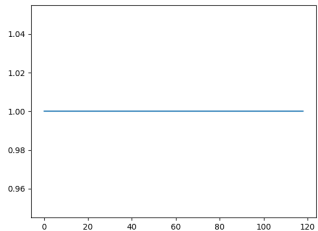
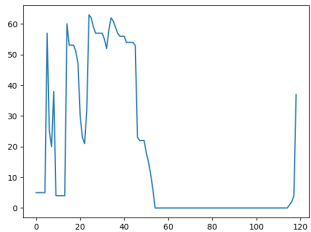
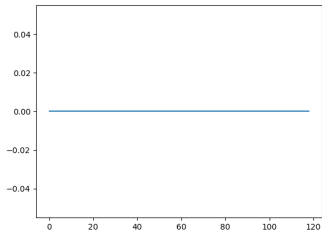


Figure: Street is viewable



Possible Improvements

- ▶ Using a Variational Auto Encoder
- ▶ Use shielding

Thanks for the attention!